

28th-09-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

Prepared by: Aminu Ibrahim Jalo
Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Tittle: BUILDING YOUR FIRST PLC PROGRAM

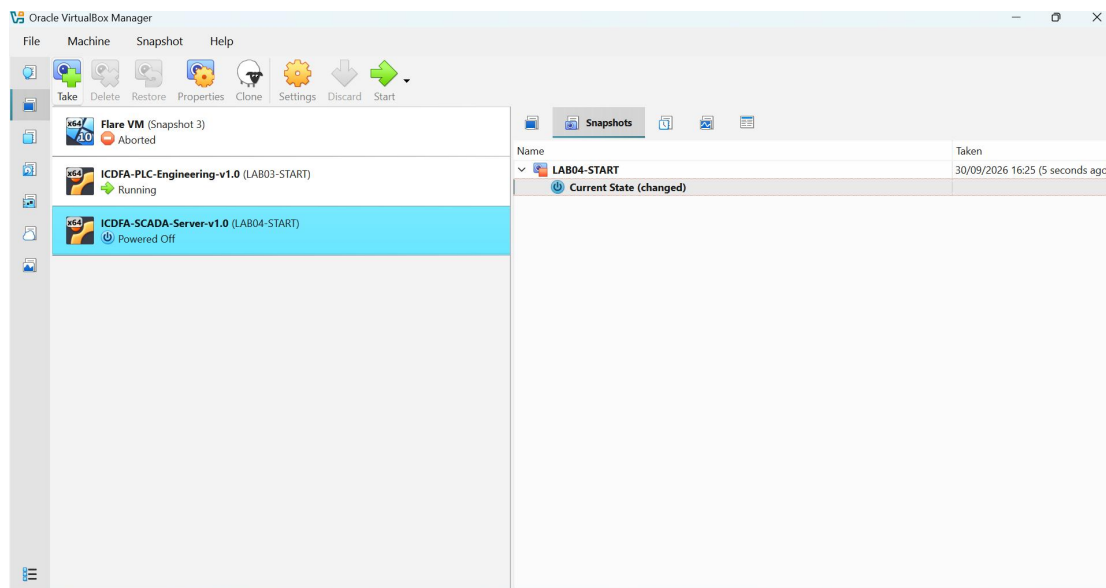
Assignment: Week 1-Lab 2 Report
C11/26/ACIS/17359

Instructor : Aminu Idris

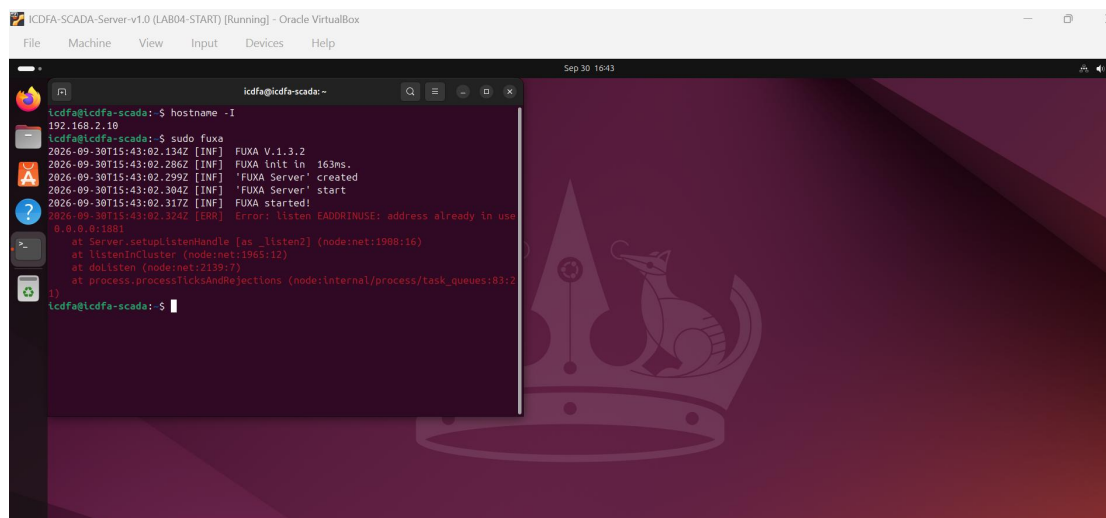
Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	28 th -09-2026

Procedure

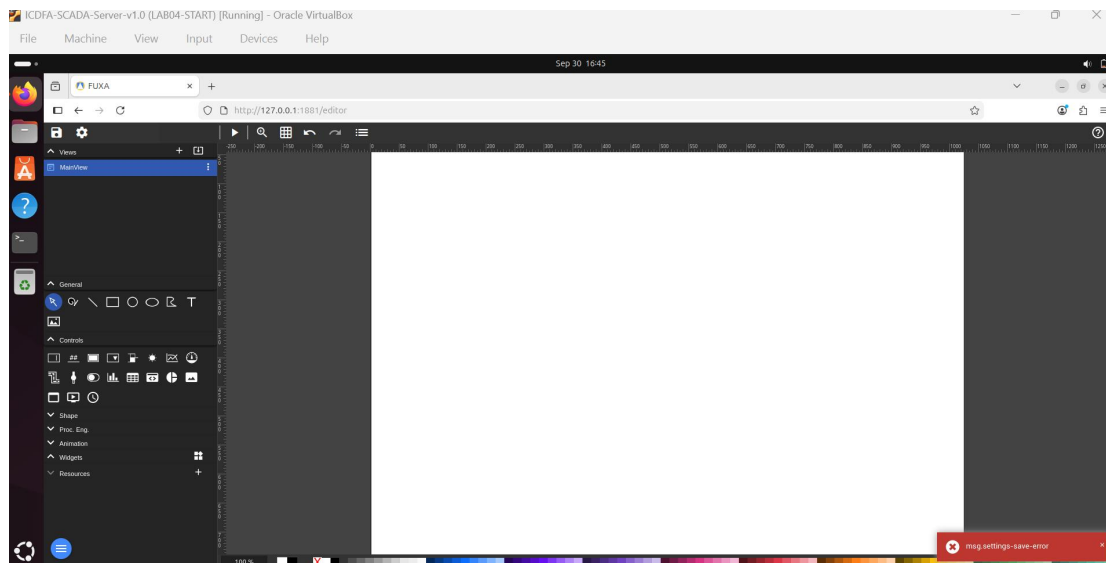
Step 1. Import/start the IC DFA PLC and SCADA VMs using the isolated network configuration supplied by your instructor. Create LAB04-START snapshots if they are not already available.



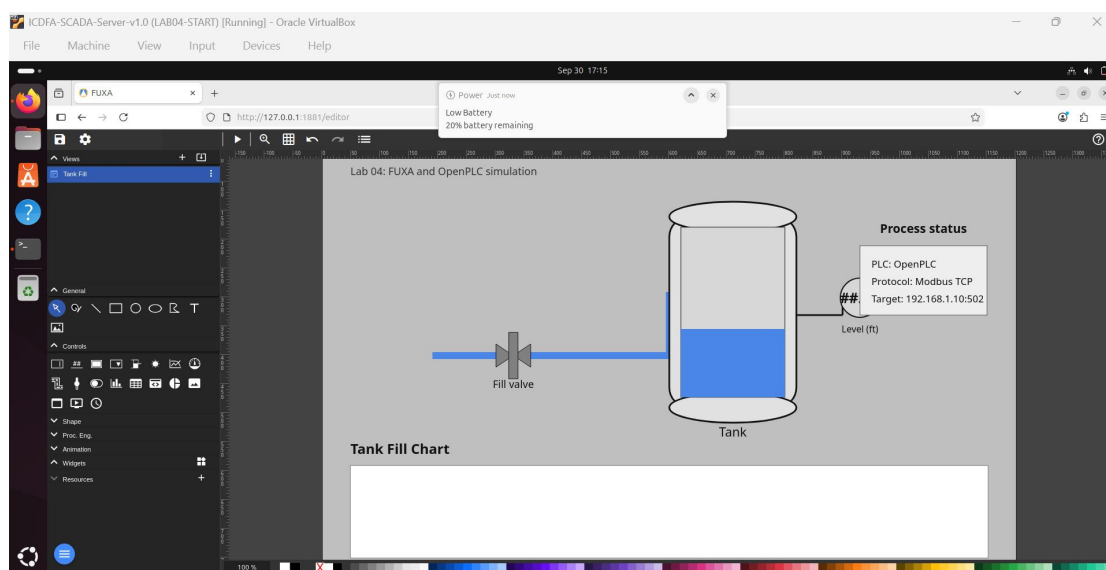
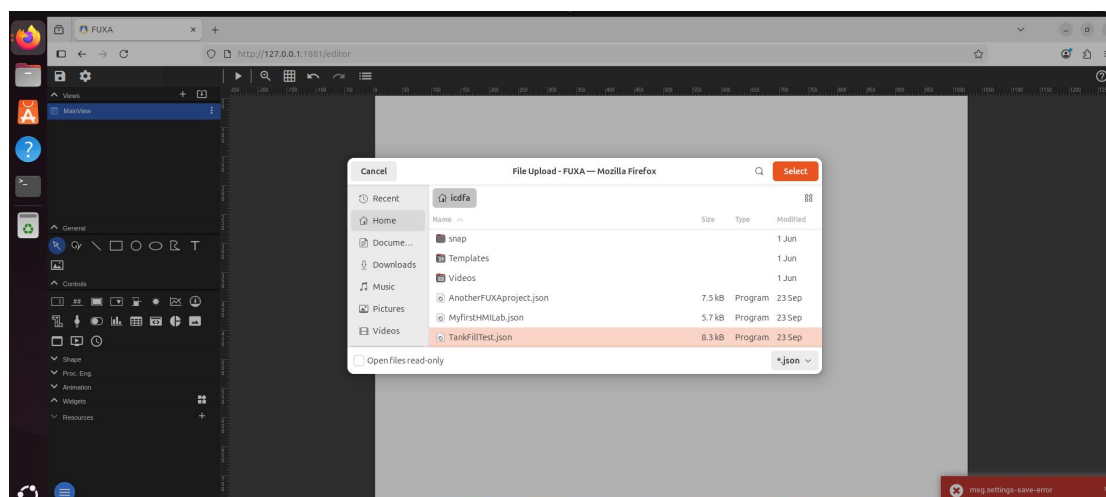
Step 2. On the SCADA VM, sign in and open a terminal. Start FUXA using the training image command: `sudo fuxa`. Enter the SCADA account password when prompted.



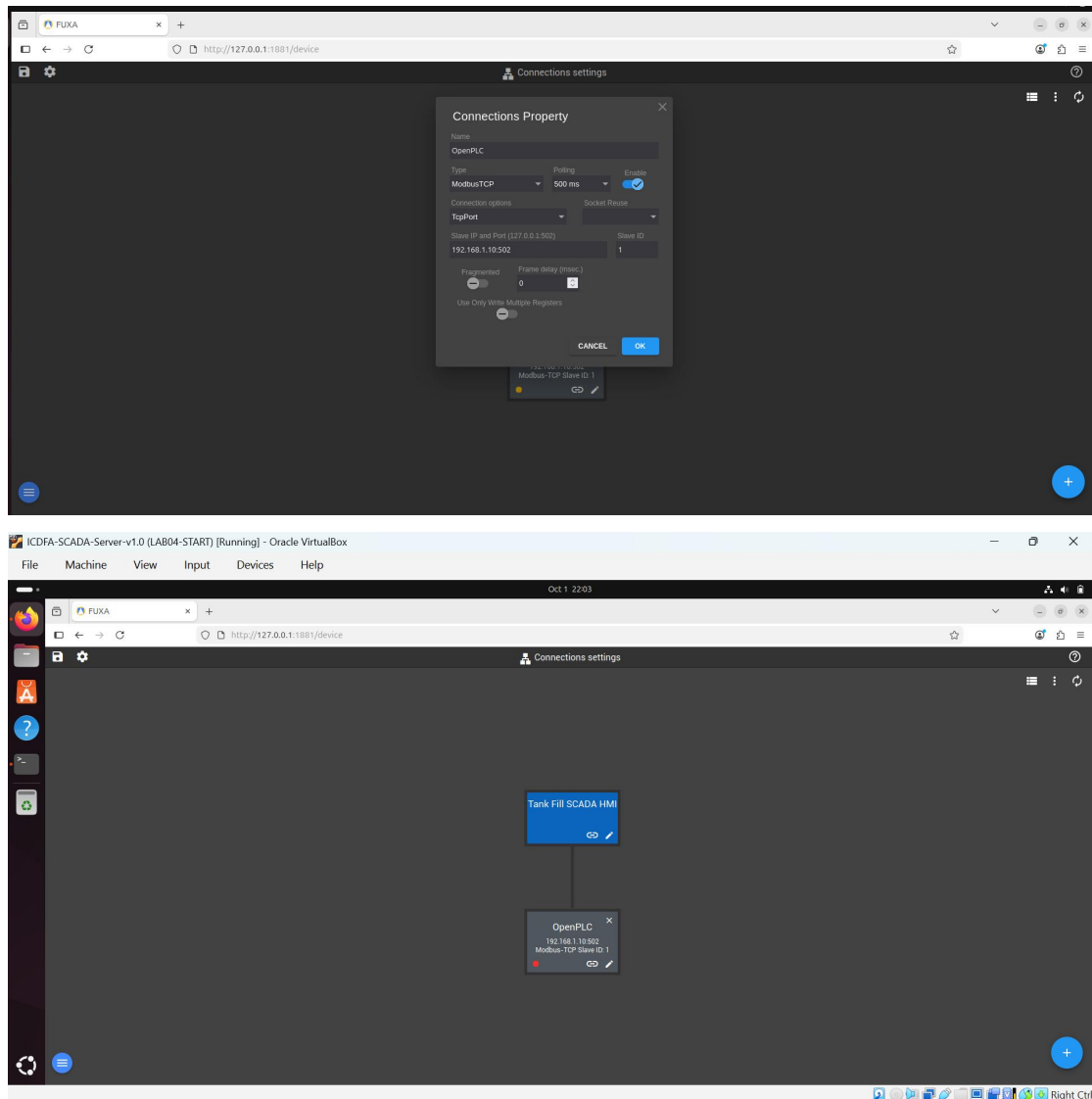
Step 3. Open Firefox and browse to <http://127.0.0.1:1881/editor> to access the local FUXA editor.



Step 4. Use the Save/Open Project function to load `TankFillTest.json` from the instructor-provided project location.



Step 5. Open FUXA project settings > Connections. Edit the openplc device and set the Modbus TCP endpoint to 192.168.1.10:502. Save the connection.



Step 6. Switch to the PLC VM. Open <http://localhost:8080/> in the local browser and sign in to OpenPLC Runtime using the credentials configured in the training image.

Step 7. Navigate to Programs, select the provided TankFillSim.st program, upload it, name it Tank Fill Sim and allow it to compile.

Step 8. Go to Dashboard and start the PLC. Confirm that the runtime status shows Running.

Step 9. Return to the SCADA VM and FUXA. Run/Play the HMI project.

Step 10. Observe the tank-fill simulation. Record the tank-level range, valve open/closed indication and any trend information presented to the operator.

Step 11. Create a simple architecture diagram in your report showing Field/Process -> PLC -> Industrial Protocol/Network -> SCADA/HMI -> Operator, and indicate where alarms/events/historian functions belong conceptually.

Step 12. Capture the required screenshots and stop the simulation cleanly when finished.

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

Response:

The HMI gives operators a clear, real-time view of the process and a safe way to control it. It turns PLC and field data into graphics, values and alarms, so operators can monitor conditions, spot problems and issue commands.

2. What is the difference between an alarm, an event record and historized process data?

Response:

Alarm: a notification that a condition needs operator attention or action, such as high tank level. It has a priority and must be acknowledged.

Event record: a time-stamped log entry of something that happened, such as an operator opening a valve, an alarm being acknowledged or a user logging in. It records activity and needs no action.

Historized process data: process values (levels, flows, temperatures) stored at regular intervals or on change, so you can trend them and analyze them later.

3. Why should a SCADA operator have context rather than only raw sensor values?

Response:

A bare number like "7.2" means little on its own. Context (units, normal range, setpoint, limits, trend, and how it relates to the equipment) tells the operator whether the value is normal, drifting or dangerous. This allows faster and more accurate decisions and reduces errors during abnormal situations.

4. What role does the PLC play compared with the SCADA application?

Response:

The PLC runs the control logic in real time at the field level. It reads inputs, executes the program and drives outputs such as valves and pumps, and it keeps working even if SCADA goes down. The SCADA application sits above it. It collects data from PLCs, displays it, handles alarms, logging and trending, and lets operators send commands and setpoints. In short, the PLC controls the process and SCADA supervises it.